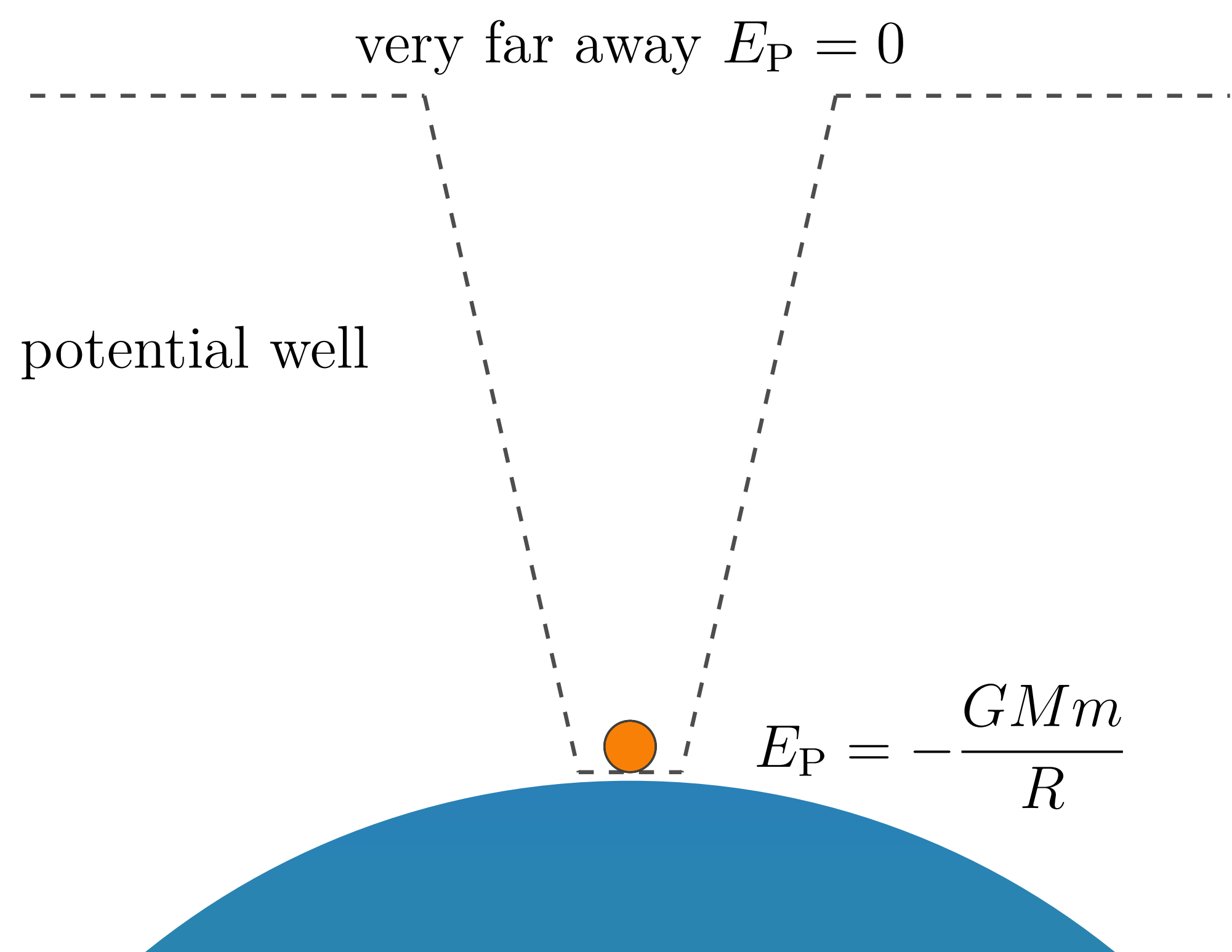
very far away $E_\mathrm{P} = 0$
potential well
$E_\mathrm{P} = -\dfrac{GMm}{R}$